Rohit Satavekar

09765371608 | Pune

rohit.satvekar@gmail.com | www.linkedin.com/in/rohitsatavekar | www.github.com/rohitsatvekar

PROFILE

AI/ML Engineer with 3+ years of experience specializing in NLP and Generative AI, with a track record of building intelligent systems for financial services, customer support, and regulatory compliance. Proficient in developing transformer-based solutions using BERT and GPT-style architectures, and implementing RAG pipelines, embeddings, and semantic search systems for real-world applications.

SKILLS

- AI/ML & GenAI: Transformers, Hugging Face, BERT, RAG, Text Summarization, Scikit-learn
- Data & Analytics: Pandas, NumPy, Power BI, Matplotlib, Seaborn, Plotly, Advanced Excel , Data Cleaning
- Backend & APIs: Python, SQL, REST API Development, FastAPI, Flask, Django
- MLOps & Data Engineering: Docker, GitHub Actions (CI/CD), Model Versioning, ETL Pipelines, Metadata Handling
- Model Evaluation: Precision, Recall, F1, AUC, FP/FN analysis, Drift detection, SHAP, LIME, A/B Testing
- NLP & LLMs: BERT, GPT, Tokenization, Embeddings, Prompt Engineering, Fine-Tuning

EXPERIENCE

• Poorit Technologies

Python Trainer *Jan 2026 – Present*

- Trained students on backend development using Python frameworks (Flask / Django / FastAPI) and frontend fundamentals along with Data Science and Analytics tools such as Pandas, NumPy, and data visualization libraries.
- Designed and delivered end-to-end Python full stack and Data Science training programs for engineering students, covering core Python, object-oriented programming, data analysis, visualization, and backend development concepts.
- Guided learners in developing full stack and data-driven mini-projects, integrating frontend, backend, databases, and analytical workflows, while following industry-standard coding practices.
- Mentored students on application deployment to cloud, including hosting, environment setup, basic DevOps workflows, and handling data pipelines for analytics applications.

• Student Loans Company

Student Finance Officer *Jan 2025 – Dec 2025*

Project I - LLM-based Customer Care Support Chatbot (Prototype)

- Built an LLM-powered chatbot prototype using RAG architecture by integrating a vector database with policy document embeddings to answer borrower queries on repayment rules, eligibility, and income thresholds, handling ~1,000 queries/day with ~25% higher accuracy than baseline FAQ responses.
- Designed intent-based query workflows and prompt templates for key use cases (repayment start conditions, income-based deductions, plan identification), and incorporated context grounding from retrieved documents to reduce hallucinations and improve factual accuracy.
- Implemented post-processing validation logic for numerical responses (e.g., repayment calculations based on income thresholds) using Python, and added guardrails (response constraints + fallback handling) to reduce hallucinated or unsupported answers, lowering critical errors by ~30%.

Project 2: Internal Policy Knowledge Assistant

- Developed an internal LLM-based assistant using LangChain for semantic search and document retrieval, reducing query resolution time by ~30%
- Processed and structured 500+ policy documents/pages using NLP techniques (chunking, embeddings) to enable context-aware retrieval and improve answer relevance by ~25%.
- Performed data preprocessing and feature engineering on 10K+ records using Pandas, NumPy, and SQL, improving dataset quality and enhancing downstream model performance by ~20%.

• Cognizant

Programmer Analyst *Oct 2021 – Oct 2022*

- Built and evaluated a face verification module for KYC onboarding using pre-trained FaceNet/ArcFace embeddings, enabling automated selfie vs ID image comparison across 10K+ image pairs.
- Generated and processed facial embeddings (128–512D vectors) and implemented cosine similarity–based matching, forming the core verification logic for identity validation.
- Tuned decision thresholds to optimize False Acceptance Rate (FAR 22%) and False Rejection Rate (FRR 18%), improving overall verification reliability.
- Performed data preprocessing and quality filtering (blur detection, normalization, resizing), improving input data consistency and boosting model robustness by ~15% on validation sets.
- Designed and analyzed model evaluation metrics, including ROC curves and confusion matrices, supporting data-driven decisions for production threshold selection.
- Conducted extensive edge-case testing (low lighting, occlusions, pose variations), identifying key failure patterns and contributing to improved model generalization.
- Integrated the face matching logic into a lightweight API/batch pipeline, enabling scalable inference with structured outputs (similarity score + decision).

PROJECTS

AI-Powered Loan Pre-Approval Chatbot (Responsible AI)

- Built an end-to-end loan pre-approval chatbot using LangChain, integrating LLM-based conversational AI with a credit risk ML model for eligibility prediction.
- Developed and evaluated an XGBoost classifier with fairness testing (Z-test, Chi-square) across protected groups to ensure unbiased decision-making.
- Applied SHAP-based explainability and bias mitigation techniques (data rebalancing, proxy analysis) for transparent and compliant outcomes.
- Implemented metadata-driven logging and governance artifacts (model cards, audit logs), reducing manual pre-screening effort by ~60%.

GenAI Regulatory Compliance Intelligence Platform | AI Consulting Project

- Designed a GenAI + RAG solution using LangChain to extract and structure regulatory obligations (FCA, GDPR, SEC) from unstructured rule-books.
- Implemented NLP-based classification aligned to risk domains (AML, Conduct Risk, Data Privacy), improving categorization accuracy and downstream analysis.
- Developed scalable document ingestion pipelines (chunking, embeddings, metadata tagging) to enable context-aware retrieval and traceable outputs.
- Reduced manual regulatory review effort by ~40% and accelerated gap analysis from weeks to days through automated, explainable GenAI workflows.

COVID-19 Detection Using Chest X-Ray Images (CNN):

- Designed and implemented a Convolutional Neural Network (CNN) using TensorFlow and Keras to classify chest X-ray images into COVID-19 positive and normal categories.
- Applied data augmentation techniques (rotation, flipping, zoom, shifting) to improve model generalization and mitigate overfitting on limited medical datasets.
- Experimented with different CNN architectures (convolutional layers, pooling, dropout, batch normalization) to optimize performance.
- Evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrix, achieving 95% accuracy and an F1-score of 0.96 on the validation dataset.

EDUCATION

• Specialized Course in Data Science & Machine Learning — *2024-2025*
Scaler

• M.Sc. Advance Computer With Data Science — *2023-2024*
University Of Strathclyde, UK

• B.E Computer Science — *2016-2021*
JSPM , Pune University